

Production lines — Screen processing design

005_DD-SPD version 1 elaborates SCR-005 and FN-032–FN-035. Order integration (FN-036 / REQ-068) is reserved to the next approved-plan step, the new order-impact DD. WI-009 plan revision 1 step 9; design only.

1. Document control and references

Field		Value	
Document ID / version		005_DD-SPD / 1	
System / subsystem		ProductionManagementAI / Master data	
Work item / author		WI-009 / Agent	
Created / updated		2026-10-01 / 2026-10-01	
Review state		Submitted for review; not implemented	

Version	Date	Author	Change
1	2026-10-01	Agent	Initial screen flows, detailed visuals and static state previews

Sources: [005 BD](#), [005 DB](#), [main DD](#), [API version 2](#), [FN version 1](#), [requirements](#), [decisions](#) and [plan](#). Approved sources remain unchanged. State diagrams/tables belong to main DD section 7; transaction algorithms belong to FN. This document owns client execution sequences.

2. Screen layout and mockup

Mockup artifact: [English static gallery](#); [Japanese edition](#). Native design/Artifact publishing is unavailable in this session; local HTML is the approved-plan fallback. No application API, write, login or interactive workflow is simulated. Captions are translated; all proposed UI remains Japanese. Each artboard is an illustrative state, not a record of a running application.

Use existing white/gray surfaces, gray borders, rounded controls, 48px inputs/actions and the Japanese system font stack from AppHeader, ProductMasterPage and index.css. Reuse shared navbar behavior; add the approved centralized Factory glyph distinct from Package/ClipboardList. At widths below 640px use menu/cards and stacked controls; at 640px and above use the shared desktop navigation/table. No new component kit.

1

ProductionManagementAI

生産ライン・工程

2

生産ライン・工程

新規ライン

3

コード・名称で検索

L-001

状態：すべて 検索 クリア

4

L-001 組立ライン A

8 時間／日 使用中

編集 使用停止

5

1 件 前へ 1 / 1 次へ

2

生産ライン編集

6

ラインコード：L-001（変更不可）

ライン名：組立ライン A

稼働時間／日：8 時間

保存 キャンセル

7

生産可能な製品 + 製品を追加

P-1001 プレーキヤリパー

製造時間／単位：2.5 分／個

使用停止予定（保存後に反映）

8

単位の再確認が必要です。

確認済み：個 現在：kg

現在の単位で確認

The mobile figure shows list and edit as two adjacent artboards for readable print review; each represents the same single-column mobile layout.

一覧

1 PMAI メニュー

2 生産ライン・工程 新規ライン

3 コード・名称で検索

L-001

状態：すべて 検索 クリア

4 L-001 組立ライン A

8 時間／日 使用中

編集 使用停止

5 1 件 前へ 1 / 1 次へ

編集

2 生産ライン編集

6 ラインコード：L-001（変更不可）

ライン名：組立ライン A

稼働時間／日：8 時間

保存 キャンセル

7 生産可能な製品 + 製品を追加

P-1001 ブレーキキャリパー

製造時間／単位：2.5 分／個

使用停止予定（保存後に反映）

8 単位の再確認が必要です。

確認済み：個 現在：kg

現在の単位で確認

9 このラインを使用停止にしますか？

L-001 / 組立ライン A

過去の製造指示は保持されます。

キャンセル

使用停止

BD region	Detail / preview states
1	Shared header; current line nav, mobile menu; dedicated Factory icon
2	List/form heading, New; dirty/retired notices
3	Unapplied q/state controls, Search/Clear
4	PC rows / SP cards; Edit and active-only line Retire
5	Loading, no lines/no matches, read error, success, paging
6	Code/name/hours, immutable edit code, validation, Save/Cancel
7	New/saved/retired timing rows, staged retirement, pair pages
8	Previous/current unit, revision-sensitive stale notice, explicit confirmation
9	Centered line/pair retirement and dirty-discard confirmations

3. Component process list

No	Owner	Responsibility / references
1	ProductionLineListPage	FN-032/035; API-PL-01/05; REQ-064/067/069
2	ProductionLineFormPage	FN-033; API-PL-02/03/04; REQ-065/066
3	TimingEditor	FN-034; API-PL-02/06; REQ-066/067
4	LineConfirmationDialog	FN-035 / staged pair action / discard; REQ-067/069
5	LineDraftValidation	Exact local validation and stable error association
6	ProductionLineApi adapter / navigation guard	Response ordering, auth, deadline, error/unknown handling

4. Processing design

4.1 ProductionLineListPage

Field	Value
Detail	Apply URL filters and handle one version-checked line retirement.
Created / last modified	Agent / 2026-10-01; initial version

Processing overview: Apply URL filters and handle one version-checked line retirement.

Used component / service	Use
AppHeader / catalog	Shared navigation, current marker and Japanese labels
ProductionLineApi / LineConfirmationDialog	List/retire and confirmation

Step	Description	Branch / condition	Calls	Result
1	Check existing session and Admin/Operator role.	401: existing login flow; 403: forbidden without rows/actions.	ProtectedRoute / auth	Loading or auth view
2	Read q/state/page from URL; initialize input draft separately.	Defaults: empty q, active, 1. Reject unknown/repeated keys or invalid bounds locally; no invalid API call.	URL parser	Valid filter draft or filter error

Step	Description	Branch / condition	Calls	Result
3	Fetch applied list; cancel superseded reads.	Accept only current request generation; old results cannot replace newer filters.	API-PL-01	Ready including empty, or ReadError
4	Search/Clear updates URL and resets page 1; paging keeps applied filters.	Clear resets q/state to empty/active. Browser history restores applied controls. Clamp navigation to available pages and 10000; no page 0.	Router / step 3	Loading; result heading receives focus after user action
5	Open New/Edit or capture active line id/code/name/version and invoker for Retire.	Retired rows have Edit only; no restore/delete.	Router / LineConfirmationDialog	Form route or region 9 overlay
6	After confirmation send one retirement using captured version.	Cancel/Escape: no request. Pending: suppress repeat. Success: announce and refresh current filters. Known failure: retain target; ambiguous: Unknown.	API-PL-05 / adapter	Retiring → Loading / Ready / Unknown

4.2 ProductionLineFormPage

Field	Value
Detail	Maintain an immutable baseline plus a draft; save one aggregate without implicit retries.
Created / last modified	Agent / 2026-10-01; initial version

Processing overview: Maintain an immutable baseline plus a draft; save one aggregate without implicit retries.

Used component / service	Use
TimingEditor / LineDraftValidation	Pair intents and validation
ProductionLineApi / navigation guard	Detail/save, errors and dirty exits

Step	Description	Branch / condition	Calls	Result
1	Authorize route; initialize create or fetch edit detail page 1.	Create: blank fields, zero pairs, Clean. Edit: no editable fake baseline on loading/error/404.	API-PL-02 for edit	Clean or local read/404 view
2	Store line version and loaded row observations; track all changes by product ID/local key.	Code read-only in edit; retired line name/hours editable without reactivation. Any changed value/action/confirmation makes Dirty.	TimingEditor / guard	Dirty; code/state locks retained
3	On Save normalize line text and validate entire draft/change ledger.	Clean/pending/Conflict/Unknown: disabled. Invalid: retain rows/values, show summary, focus first invalid control; no mutation.	LineDraftValidation	Dirty errors or Saving

Step	Description	Branch / condition	Calls	Result
4	Build create products or edit productChanges; capture submitted index-to-row map.	Edit sends only changed actions, version, name/hours; no code. Omitted rows unchanged. Reject > 1000 actions or >256 KiB locally without silent truncation; persisted pair count unlimited by this request cap.	API-PL-03 / API-PL-04	One pending aggregate request
5	Handle confirmed response or failure through adapter.	Success: navigate list with one announced notice. Validation: map captured indices to stable rows. Stale parent/unit: Conflict. Ambiguous: Unknown; never clear draft early.	Adapter / router	Exit / Dirty / Conflict / Unknown
6	Resolve Conflict/Unknown explicitly; guard replacement of dirty data.	Edit verify by id; create verify by list/code. Matching code alone does not prove own creation. If outcome cannot be reconciled keep write disabled; no replay/auto merge.	API-PL-02 / API-PL-01 / discard dialog	Read verification and explicit authoritative reload

4.3 TimingEditor

Field	Value
Detail	Keep a cross-page action ledger and explicit observed-unit confirmation.
Created / last modified	Agent / 2026-10-01; initial version

Processing overview: Keep a cross-page action ledger and explicit observed-unit confirmation.

Used component / service	Use
ProductionLineApi / decimal validator	Product choices, paged detail and exact minutes
LineConfirmationDialog / form draft	Stage retirement and track changes

Step	Description	Branch / condition	Calls	Result
1	Fetch product choices by applied search/page and lineld for edit; exclude local unsaved IDs.	Server excludes every saved association including retired. New choice must be active; no duplicate product IDs. Cancel obsolete reads.	API-PL-06	Choices or retryable read error
2	Add a selected product observation with blank timing and unconfirmed intent.	No automatic conversion/confirmation; require minutes per one displayed unit. Remove unsaved row locally with no API.	Form draft	Dirty new row
3	Read saved pair pages into keyed baseline cache; keep intents even off page.	Compare returned parent version to original. Changed version: Conflict, guarded reload; never merge. Same version: update current observations and clear intent if unit/revision changed.	API-PL-02 with pairsPage	Current page plus preserved action ledger

Step	Description	Branch / condition	Calls	Result
4	Edit active saved minutes or explicitly confirm displayed unit/revision.	Stale pair needs confirmation even if unit text unchanged. New pair requires confirmation. Fresh timing edit may retain current confirmation (confirmUnit false). Later coefficient/observation change clears intent.	Exact validator / form draft	setTiming/add with observed unit and opaque revision
5	Keep unchanged stale/inactive-product associations during unrelated edits.	Saved retired pair read-only. No add/reactivation of its product. Stale edited timing requires confirm; eligibility remains server owned.	Form draft	History retained; no implicit action
6	Confirm saved active pair retirement and replace any timing intent with retire action.	Only stages local intent; expose pending-retirement text. Save persists; Cancel/discard writes nothing.	LineConfirmationDialog / form draft	Dirty pending retirement

4.4 LineConfirmationDialog

Field	Value
Detail	Center a safe modal and distinguish immediate line retirement from staged pair retirement and discard.
Created / last modified	Agent / 2026-10-01; initial version

Processing overview: Center a safe modal and distinguish immediate line retirement from staged pair retirement and discard.

Used component / service	Use
Native dialog / remembered invoker	Top layer, keyboard containment, safe and return focus
List/form action / navigation guard	Dispatch exactly the approved action

Step	Description	Branch / condition	Calls	Result
1	Capture target text, operation and invoker; call showModal().	Never derive action from untrusted HTML; text rendering only. Associate heading/body with aria-labelledby/describedby.	Native dialog	Overlay, inert background
2	Apply viewport geometry and focus safe action.	position fixed; inset 0; margin auto; width min(480px, calc(100vw - 32px)); max-height calc(100dvh - 32px); overflow auto. Center after scroll, SP and zoom.	Cancel / keep editing focus	Both axes centered; no page-relative top offset
3	Cancel, Escape or confirm.	Cancel/Escape closes with no write or draft change. Confirm line → API-PL-05 once; confirm pair → local retire; confirm discard → requested exit/reload only.	Owner action	Remembered return state or pending/exit
4	Close and restore focus.	Invoker if still present; after refreshed retirement use row/status heading; completed navigation uses destination heading. Backdrop click is not confirmation.	Owner focus target	Predictable keyboard position

4.5 LineDraftValidation

Field	Value
Detail	Validate exact text without floating-point rounding; preserve field identity.
Created / last modified	Agent / 2026-10-01; initial version

Processing overview: Validate exact text without floating-point rounding; preserve field identity.

Used component / service	Use
API v2 grammar / draft ledger	Format, bounds, action identity and confirmation

Step	Description	Branch / condition	Calls	Result
1	Trim code/name only; count Unicode code points.	Create code 1–50, name 1–200. Edit code untouched. Server DB lower(code) governs duplicate checks across retired lines.	Local validator	Field errors; never preclaim uniqueness
2	Validate hours/minutes as ASCII text with API regex.	^(0 [1-9][0-9]*)(.[0-9]{1,3})?\$\$; hours >0 <=24; minutes >0 <=999999999.999. No spaces, signs, leading zeros, exponent, grouping or precision rounding; 1.2340 invalid. Compare exact scaled decimal values; never Number.	Exact decimal helper	Retained text or field error
3	Validate each action and explicit observed-unit intent.	One action/product. add requires confirmUnit true; stale setTiming requires true; retire contains action/productId only. Unit in 個, 本, 枚, 台, セット, kg, m. Opaque versions compared as strings.	Action ledger	Valid aggregate or stable-key errors
4	Map server errors through captured submission order.	products[i] / productChanges[i] map to row even off page. Show page/row error link and reveal before focus; unknown field paths become summary, never dropped. Server remains authoritative.	Error summary / form	All draft values retained

4.6 ProductionLineApi adapter / navigation guard

Field	Value
Detail	Handle cancellation, known rejection and unknown writes without leaking or storing drafts.
Created / last modified	Agent / 2026-10-01; initial version

Processing overview: Handle cancellation, known rejection and unknown writes without leaking or storing drafts.

Used component / service	Use
Existing apiClient / auth / navigation guard	Same-origin session and guarded navigation

Step	Description	Branch / condition	Calls	Result
1	Send same-origin JSON with cookie; validate decoded representation.	Reads generation checked; write deadline 20s. Abort/network/lost response does not prove rollback. No automatic mutation retries.	API-PL-01-06	Decoded result or classified failure
2	Classify application response and keep local data.	400 / code conflict: field/summary error; LINE_STALE, LINE_UNIT_STALE, LINE_UNIT_CONFIRMATION_REQUIRED: Conflict. 404 local missing view. 413/415 actionable summary. 503 LINE_BUSY is known rollback; explicit retry only, honor Retry-After. 500/malformed/lost write → Unknown.	Problem code mapping	Dirty / Conflict / Unknown / read error
3	Apply existing 401 session handling; stop feature access on 403.	Do not expose protected baseline on forbidden routes; drafts memory only and cleared by existing logout/expiry. No localStorage, URL payload or new persistence.	Auth / ProtectedRoute	Login or forbidden
4	Guard Cancel, navbar and browser navigation/reload while dirty; suppress exits during pending writes.	In-app uses discard dialog; reload/unload uses native browser prompt without custom translation guarantees. Conflict/Unknown verification reload also guards discarded data.	Existing navigation guard / beforeunload	Retain or explicitly discard; never implicit save
5	Emit only bounded client operation/status diagnostics if existing instrumentation supports it.	No bodies, q, names, coefficients, unit/version tokens or credentials; new server spans/metrics remain FN owned.	Existing diagnostics	No new telemetry dependency

5. UI messages and accessibility

New feature labels below are catalog proposals; existing shared labels must match messages.ts exactly. Centralize new text under the same catalog when implementation is approved. Wireframes/mockups show Japanese only; captions are outside the UI.

Context	Japanese UI	English gloss
Heading	生産ライン・工程	Production lines and processes
Create / edit	生産ライン登録 / 生産ライン編集	Create / edit production line
New / supported products	新規ライン / 生産可能な製品	New line / supported products
Identity / hours	ラインコード / ライン名 / 稼働時間／日	Code / name / working hours per day
Timing / confirm	製造時間／単位 / 現在の単位で確認	Minutes per unit / confirm for current unit
Stale / staged retirement	単位の再確認が必要です。 / 使用停止予定（保存後に反映）	Unit reconfirmation required / retirement pending Save
Loading / empty / no match	読み込み中... / 生産ラインはまだ登録されていません。 / 該当する生産ラインはありません。	Loading / no registered lines / no matches
Read error / retry	読み込みに失敗しました。 / 再試行	Read failed / retry
Saved / retired	生産ラインを保存しました。 / 生産ラインを使用停止にしました。	Line saved / retired

Context	Japanese UI	English gloss
Conflict / unknown	変更が競合しています。 / 保存結果を確認できません。再送信せず、現在の状態を確認してください。	Conflicting change / unknown save; verify without replay
Forbidden	この機能を利用する権限がありません。	No permission for this feature

WCAG 2.2 AA target: label every input with persistent required/unit hints; associate field errors with aria-describedby and aria-invalid; error summary links reveal paged rows. Table has caption and scoped headers; cards use named regions. Status is text, not color alone. Use polite live status for loading/success and an alert for errors, without duplicate announcements. DOM focus order: header → heading/New → filters/results/pages, or identity → pair actions/pages → Save/Cancel. Preserve visible focus rings, 48px controls, sufficient contrast, reflow at 320px and 200% zoom. Dialog keyboard containment, inert background, Escape, focus restoration and scroll-safe viewport centering require later runtime verification; static previews cannot prove them.

6. Verification viewpoints and review stop

Requirement	Later verification
REQ-064	URL history; literal q; state/page bounds; read cancellation; table/cards; two empty states
REQ-065	Immutable code; trim/case duplicate; exact text limits; version conflict retains draft
REQ-066	Numeric boundaries; fresh/stale/ABA confirmation; unit-change race; cross-page intents/errors
REQ-067	Cancel/Escape no write; pair staged vs line immediate; history/read-only retired; no restore
REQ-068	Covered later by new order-impact DD; no new order screen authored here
REQ-069	Admin/Operator read/write; 401/403; native modal focus/geometry; keyboard/mobile/zoom

Business decisions unresolved: none. Source/API constraints are settled by approved inputs. Current checks cover document/visual consistency only; no application, network, accessibility conformance, transaction or E2E result is claimed. Detailed actual checks are recorded in [evidence](#). Submit this single design package and wait for explicit user review before writing 005_DD-ORD. Full-family/security reconciliation and implementation plan revision 2 remain later gates.